ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S

Unit 7.01: Systems of Linear ODEs & Phase Plane Topology

Name: _____

Q1. The matrix of mathbf x'=Amathbf x has eigenvalues pm i . The origin is a:

[ ] A: saddle point unstable
[ ] B: unstable spiral
[ ] C: stable node
[ ] D: center stable, not asymptotically

Q2. Solve x'=2 x, y'=- y with x0=-1, y0=3 . Find xln{left2 right} .

[ ] A: - frac{1}{2}
[ ] B: 4
[ ] C: 12
[ ] D: -4

Q3. With x_1=y , x_2=y' , the equation y''+3y=0 becomes mathbf x'=Amathbf x . Find A .

[ ] A: left[begin{matrix}0 & 13 & 0end{matr
[ ] B: left[begin{matrix}0 & 1-3 & 0end{mat
[ ] C: left[begin{matrix}1 & 0-3 & 0end{mat
[ ] D: left[begin{matrix}0 & 10 & -3end{mat

Q4. The matrix of mathbf x'=Amathbf x has eigenvalues pm 5i . The origin is a:

[ ] A: center stable, not asymptotically
[ ] B: unstable node
[ ] C: saddle point unstable
[ ] D: unstable spiral

Q5. Solve x'=- x, y'=- y with x0=-2, y0=4 . Find yln{left2 right} .

[ ] A: -2
[ ] B: 2
[ ] C: -1
[ ] D: 4

Q6. With x_1=y , x_2=y' , the equation y''+y'+y=0 becomes mathbf x'=Amathbf x . Find A .

[ ] A: left[begin{matrix}0 & 1-1 & -1end{ma
[ ] B: left[begin{matrix}0 & 11 & 1end{matr
[ ] C: left[begin{matrix}1 & 0-1 & -1end{ma
[ ] D: left[begin{matrix}-1 & -11 & 0end{ma

Q7. The matrix of mathbf x'=Amathbf x has eigenvalues -2pm 4i . The origin is a:

[ ] A: stable spiral
[ ] B: unstable spiral
[ ] C: unstable node
[ ] D: stable node

Q8. Solve x'=- x, y'=y with x0=4, y0=-1 . Find x1 .

[ ] A: 4 e
[ ] B: - frac{4}{e}
[ ] C: - frac{1}{e}
[ ] D: frac{4}{e}